

Mano Bhavdeep Venkatesan

Biomedical & Microsystems Engineer | Thin Films & Surface Characterization (ALN, ALD, Sputtering)

Biomedical & materials engineer specializing in **Aluminum Nitride (ALN) thin films** on Si/SiO₂/Poly-Si MEMS structures for acoustic sensing. Skilled in FT-IR, XRD, SEM and wafer bow for stress, texture and morphology, plus Fresnel-based MATLAB simulations and Brings 3+ years of hospital/clinical equipment engineering background into **materials-driven R&D**, clinical equipment QA in medical technology and microsystems.

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WORK EXPERIENCE

MSc Master Thesis - Acoustic MEMS Research Hahn Schickard Gesellschaft

04/2025 - 10/2025

Villingen Schwenningen

Development and characterization of 3D membranes using FT-IR Spectroscopy for acoustic applications. **Grade 1.0**

Achievements/Tasks

- Developed reflection-mode FT-IR protocols on **ALN-coated 3D MEMS membranes** (Si/SiO₂/Poly-Si, trench widths 10–80 µm) to extract phonon modes for residual stress and crystal orientation analysis on vertical sidewalls.
- Established a 90° sidewall mounting and aperture optimization workflow on a **Bruker Hyperion 2000** microscope to reproducibly measure ALN sidewall spectra.
- Correlated FT-IR phonon peak shifts with **XRD θ –2 θ and rocking curves** (Bruker D8 Advance), obtaining c-lattice, crystallite size, and texture ratio for c-axis oriented ALN.
- Compared **ALD vs RF magnetron sputtered ALN** films: quantified trade-off between ALD sidewall conformality and sputtered (002) texture/non-uniform coverage using SEM cross-sections (Zeiss Crossbeam 550), wafer bow measurements (E&H MX203), and Fresnel-based MATLAB reflectance simulations.

Biomedical Engineer

SIMATS (Saveetha University, India)

01/2019 - 01/2022

Chennai, India

<https://www.saveetha.com>, <https://saveethadental.com>

Achievements/Tasks

- Operated and maintained **SEM and INSTRON tensile/compression testers** for routine failure analysis and mechanical characterization of medical device components and biomaterials, ensuring 99% lab uptime in a high-throughput clinical environment.
- Implemented QA workflows aligned with **NABH and ISO-based safety standards**, reducing equipment audit non-conformities by improving calibration, documentation, and preventive maintenance routines.
- Installed, calibrated, and validated > 50 diagnostic and monitoring systems (imaging, cardiology, and critical-care devices), ensuring compliance with performance specifications and safety protocols.

EDUCATION

M.Sc. Biomedical Engineering

Hochschule Furtwangen University

Villingen-Schwenningen, Germany

B.E Biomedical Engineering

Jerusalem College of Engineering

Chennai, India

SKILLS

Laboratory & Material Characterization

Regulatory & Quality Assurance

Engineering & Design

Data Science & Simulation

Machine Learning & Data Analysis

Python, MATLAB

RESEARCH PROJECTS

Detection of Out of Body Images in Laparoscopic Videos

- Implemented MATLAB pipelines for detecting out-of-body frames in laparoscopic videos using supervised classifiers (SVM, decision trees, neural network models) on features derived from color histograms, motion cues, and specular highlights. Evaluated models on labeled surgical datasets to improve tool-in-body frame detection and reduce surgeon distraction.

Automated Detection of Maculopathy in SD-OCT Images (Bachelor Thesis)

- Developed a MATLAB pipeline for automated maculopathy screening in SD-OCT images targeting diabetic retinopathy, diabetic macular edema, and AMD.
- Segmented retinal layers, extracted statistical and texture features, and trained a multilayer SVM classifier achieving 93% accuracy, 80% sensitivity, and 100% specificity on labeled OCT datasets.

CERTIFICATES

Medical Device Regulation (EU MDR), Python Fundamentals, Biomaterials for Bioprinting – Udemy

Six Sigma Green Belt - LinkedIn Learning course

LANGUAGES

English

Native or Bilingual Proficiency

Tamil

Native or Bilingual Proficiency

German

Professional Working Proficiency

INTERESTS

MEMS Sensors - Acoustic/ultrasonic

Biointerface engineering & osseointegration

Smart Implants & Bio-Interfaces

Materials characterization

Surface & materials analytics

R&D from lab to product